

Diagnosis and Treatment of Root Caries

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Evidence indicates that root caries was the major dental problem in ancient civilizations.[1,2] Today, demographic predictors suggest that root caries could become one of the more significant patient management issues of the future.[3–9] Greater life expectancies coupled with improved dental care have resulted in an increasing number of patients who retain most of their dentition into old age. About 75% of the US population aged 60 years or older are dentate (mean, 18.9 teeth).[10] With age, the occurrence of root surfaces exposed to the oral environment increases, predisposing the affected teeth to loss of cervical structure, cervical hypersensitivity, and root caries.[11] In a study assessing the periodontal status in the United States,[12] gingival recession of 1 mm or more was reported in 11.5% of 18- to 24-year-olds, 46.3% of 35- to 44-year-olds, 78.3% of 55- to 64-year-olds, and 86.5% of people aged 65 years or older. Consequently, the risk for root caries increases in the elderly, dentate patient.[8,13–18]

Definition, Clinical Appearance, and Location of Root Caries Lesions

The prevalent definition in the dental literature for a *root caries lesion* is "a soft, irregularly shaped lesion either *(1)* totally confined to the root surface or *(2)* involving the undermining of enamel at the cementoenamel junction, but clinically indicating that the lesion initiated on the root surface."[19] A root caries lesion can be initiated only if the root surface is first exposed to the oral environment.[7,12,20,21] A root caries lesion appears as a softening and/or cavitation in the root surface with no initial involvement of the adjacent enamel (Fig 13-1). These lesions generally begin at or slightly occlusal to the free gingival margin where dental plaque more frequently accumulates undisturbed but can extend into the gingival sulcus and/or undermine the coronal enamel as the lesion progresses (Fig 13-2). Lesions also begin at the margins of restorations that have their cervical interfaces on root structure. Two reports by Mjör[22,23] indicate that secondary caries lesions occur more frequently at cervical margins because many restorations terminate on root surfaces in areas where access and isolation are most difficult. Because dental plaque frequently accumulates at the gingival margin around the tooth, an active root caries lesion, if left untreated, can spread laterally, encircling the tooth[24] (Fig 13-3).



Fig 13-1 Root caries lesion on a tooth with gingival recession.



Fig 13-2 Root caries lesion undermining coronal enamel.



Fig 13-3 Active root caries lesion extending laterally.

Early-stage lesions can be difficult to diagnose by appearance, because color changes are frequently not obvious until some progression of the caries lesion has occurred. New lesions may appear as small, well-defined areas of a yellowish to light brown color. On probing, the dentin in an active lesion is softer than adjacent, unaffected cementum or dentin and can be removed with a sharp excavator. Advanced lesions appear darker brown to black and, if arrested, may be as hard or harder than the normal root surface. There has been an attempt by researchers to categorize lesions based on color (lighter lesions more active, darker lesions inactive) and/or texture (the harder the lesion, the less active).[25–28] While there has been some relationship shown between color, texture, and dominant microorganism, the data have been conflicting and the link remains tenuous.[29] Currently, color is not considered a reliable indicator of caries activity.[17,30–32]

Caries lesions may occur on any exposed root surface, but initial lesions on the facial and proximal surfaces are most common.[33] Some studies have suggested that 50% to 75% of root caries lesions begin proximally[34,35] (Fig 13-4). This increased incidence of root caries is likely due to limited access to the cleansing forces of the oral cavity (eg, tongue) and the loss of the protective interproximal tissue, allowing increased food and bacterial retention. Lingual/palatal locations are seen much less frequently as isolated lesions (Fig 13-5). In the mandible, molars appear to be the most susceptible to root caries, followed by premolars, canines, and incisors; in the maxilla, the order is reversed.[33,36,37] It is common for many of these lesions to be obscured by plaque and food debris, so accurate detection is best accomplished after thorough debridement and prophylaxis.

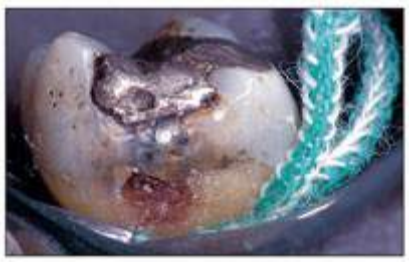


Fig 13-4 Proximal root caries lesion on a second molar.



Fig 13-5 Caries lesion on the lingual aspect of a second premolar.

Histochemistry, Histopathology, and Microbiology

The caries process on root surfaces is very similar to that in coronal caries. Plaque bacteria capable of metabolizing dietary carbohydrates into acids produce a drop in pH that lowers the plaque fluid saturation, initiating demineralization of the tooth structure. Saturation in plaque fluid, particularly near the tooth crystals, is what determines if tooth crystals demineralize, remineralize, or remain stable. It is the physicochemical basis for the caries process. For example, low pH (eg, 4.8) at higher levels of saturation will not demineralize enamel. Root surfaces are more vulnerable to chemical dissolution than enamel surfaces.[38] The drop in pH to undersaturate the biofilm in respect to cementum and dentin is much less than that required for enamel.[39–41] This means that, given the proper environment, both the initiation and progression of root surface caries lesions will occur more rapidly in dentin than in enamel.[42] Denaturation of collagen by collagenase-producing microorganisms, as well as degradation by nonspecific proteases, may accelerate the problem.[43] In addition, acid challenges can occur more readily and may continue for an extended period of time.[44] Frequent alterations in the delicate balance between the rates of demineralization and remineralization can result in the initiation of the caries process.[45–47]

While we tend to think of root surfaces as being covered with cementum, it has been shown that the cementum and enamel are not confluent in as many as 30% of teeth.[48] For patients receiving periodontal therapy, the cementum on accessible root surfaces is often partially removed during scaling and root planing procedures. Therefore, root caries lesions commonly begin on a dentin interface. Regardless of the surface, the creation of an acidic environment by cariogenic bacteria initiates the caries process. Cemental clefts can form due to physical and chemical changes, allowing infiltration of bacteria into the dentinal tubules. Surface dissolution

continues, followed by further demineralization and destruction of the collagen matrix.[49,50] Early microcavitation enlarges and produces the characteristic circumferential spreading seen with these lesions[50,51] (Fig 13-6).

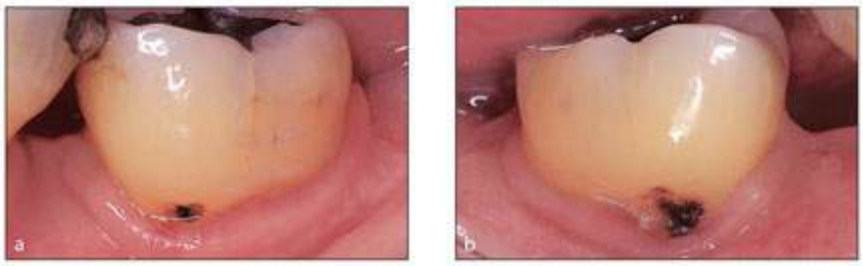


Fig 13-6*(a)* Small lesion in a 45-year-old patient. At this stage, if the lesion is active (soft), with cavitation, a restoration should be placed to prevent circumferential spread, and a preventive regimen should be initiated. *(b)* Larger lesion in the same patient.

As demineralization progresses, there is a reactive sclerosing of the tubules and crystal formation, resulting in hyperminer-alization of the dentinal tubules.[52] This sclerosis is believed to be a result of the pulp's reaction to the stimulus of the caries process. The sclerotic appearance of many advanced lesions is probably related to the differences in mineral content found between the peritubular and intertubular dentin.[52,53] However, lesions can become arrested when conditions at the surface of the lesion change, leading to remineralization. Preventive regimens can lead to the arrest of root surface lesions. Arrested lesions frequently appear very dark and are hard on tactile examination.[54]

Many studies attempting to identify the etiology of root caries have used selective culture techniques that focus on the identification of a limited number of bacterial species.[55,56] Unfortunately, this type of culturing often excludes many species that may be directly or indirectly related. Studies of the predominant cultivable microflora as identified by microarrays, while more time and labor intensive, appear to be more useful in delineating the group of oral bacteria associated with the root caries process.[46,57–59]

Currently, no specific single microorganism has been conclusively proven to cause root surface caries. Early studies[28,60,61] pointed to *Actinomyces viscosus* as a prime suspect. More recent studies, however, have questioned the dominant role of *Actinomyces* while emphasizing the importance of mutans streptococci and lactobacilli.[25,29,59,62–68] Lynch[29] and Zambon and Kasprzak[46] point out that, based on extensive research, *Streptococcus mutans* and *Lactobacillus* both fulfill the criteria for implicating bacteria in the etiology of a mixed infection. They also suggest that virulence factors of specific species probably play an important role in both the

formation and progression of root surface lesions. It is likely that root caries is a continuous, destructive process involving a succession of bacterial populations that vary depending on the condition of the substrate and the depth of the lesion.[57,67] Modern molecular biology techniques, involving ribosomal RNA sequencing and DNA and RNA probes, may offer solutions to defining both species and virulence factors associated with root caries.[59,69]

Prevalence and Incidence

Because root caries lesions can be initiated only when root surfaces are exposed to the oral environment, the population presumed to be most at risk are older adults. However, younger patients with periodontal problems are susceptible as well. The prevalence of 1 mm or more recession in people aged 30 years and older in the United States has been estimated at 58%, increasing with age. About 24 million persons in the United States are expected to have one or more tooth surfaces with recession of 3 mm or greater.[70] It also should be noted that root surface exposure does not mean that caries activity is inevitable. The actual prevalence of root caries is difficult to assess.[17,31,71]

Interpretation of data from prevalence and incidence studies is complicated by differences in diagnostic criteria, treatment decisions, and lack of homogeneity of the observed populations.[72] Numerous studies have reported the prevalence of root caries and its relationship to increasing age,[10,14,33,73–75] while international surveys have estimated that the disease affects a significant percentage of adults.[11,27,76–78] In addition, it has been suggested that approximately one in nine root surfaces is at risk of becoming carious.[33] The extent of root caries appears to have a negative correlation with the number of teeth present. Thus, the ongoing loss of teeth with age is likely to produce an underestimation of the prevalence of root caries.[14] Because more mandibular teeth are retained in older individuals, these teeth have a higher incidence of root caries lesions than do maxillary teeth.[33,79–81] Other studies[33,76,82–85] suggest that approximately 15% to 20% of all teeth with gingival recession are affected by root caries, and the mean number of teeth affected per person is about 2.8.[38,77,86–88] It has also been stated that if root caries prevalence is based on the presence of active, restored, and arrested lesions, virtually every dentate person older than 65 years of age is at risk.[88]

Incidence data have been derived primarily from studies conducted on selected populations, such as the chronically ill or nursing home residents. These

studies[37,74,89–92] vary in duration from 1 to 8 years and report root caries/root restoration experience ranging from 19% to 69% depending on the population observed. Two studies on noninstitutionalized adults older than 65 years of age have reported similar incidences of root caries, 44%[93] and 37%.[36] The attack rates (mean carious surfaces per mouth) calculated for exposed root surfaces in the latter two studies ranged from 3.9 to 5.5 over a 3-year period.

Despite the variability of available data, there was a general agreement that the prevalence of root caries will increase in the dentate older population as a result of the population retaining their teeth longer. However, the latest large-scale surveys in the United States suggest that root caries prevalence is declining despite an increase in tooth retention and no change in coronal caries. The overall prevalence of root caries for all adults decreased from 19% to 14%, with the greatest decline in adults aged 50 to 64 years (9%) and older adults living at the lowest level of income (16%).[10]

Risk Factors and Assessment

The risk factors associated with root caries are provided in Box 13-1. It is of critical importance that clinicians identify at-risk patients early in the root caries process, ideally before the disease is clinically apparent. Early detection permits preventive and chemotherapeutic intervention to potentially enhance treatment outcome.

Box 13-1	**Risk factors for root caries**
Exposure of root surfaces Attachment loss Gingival recession Periodontal pocketing Inadequate oral hygiene Low priority to patient Physical impairment Cognitive impairment Cariogenic diet Diminished salivary flow and/or buffering capacity Chronic medical conditions Medications Surgical/radiation therapy Physiologic aging Previous caries lesions/restorations Lack of access to and/or interest in dental services Low socioeconomic status	

Low educational level
Removable prosthesis
Advanced age
Eight or more missing teeth
Male sex
Smoking, alcoholism, drug use
Possibly certain ethnicities

Because exposure of the root surface to the oral environment is a prerequisite for root surface caries, any patient with attachment loss, gingival recession, and/or periodontal pockets is at risk for initiation of the disease process.[94] Patients in this category who are frequently overlooked are patients with cervical and proximal restorations that terminate on cemental surfaces. Even though the root surface may not be readily visible, the need for and placement of these restorations has met the primary risk criteria.

All normal risk factors for caries lesion development are applicable to root caries, including inadequate oral hygiene, cariogenic diet, and poor utilization of routine dental services.[95,96] Past caries/restorative experience also has been shown to have a strong correlation and generally indicates the presence of conditions/behaviors that support caries activity[36,87,97–104] (Fig 13-7). Unfortunately, the effect of these conditions may be magnified in the root caries process as well as impacted by the myriad changes associated with aging and related health problems and treatments.[31,105]



Fig 13-7 Root caries lesion adjacent to an amalgam margin.

In relation to root caries activity, salivary flow rate is considered the most important of the nonmicrobial salivary parameters,[106–108] because the cariostatic activity or efficacy of other salivary parameters is dependent on the flow rate.[109] Unstimulated flow rate has been shown to have a greater effect on salivary clearance time than stimulated flow[110,111] and is more affected by conditions producing hypofunction of the salivary glands.[112] Loss or significant reduction of unstimulated salivary flow can result in xerostomia, or "dry mouth," and is positively correlated

with a number of adverse oral conditions, including rapidly progressive caries lesions and periodontal disease[113–115] (Fig 13-8). While there is debate as to the amount of saliva necessary to maintain oral health, an unstimulated flow rate of less than 0.1 mL/min is considered to be below normal.[112–117]



Fig 13-8 Caries lesion adjacent to a resin restoration in a xerostomic patient. Glass-ionomer restorations in the same patient did not exhibit recurrent caries lesions for the duration of the 5-year study.[77]

Hyposalivation can be caused by a variety of factors,[13,39,113] including radiation therapy of the head and neck, immuno-suppressive therapy, radioactive iodine therapy, autoimmune diseases, HIV infection, and a myriad of commonly prescribed medications (Box 13-2). Basic management of patients with hyposalivation involves finding ways to reduce their oral dryness. If functioning salivary gland tissue is present, stimulation of natural flow is preferable to saliva substitutes. Pilocarpine (Salagen, MGI Pharma) and cevimeline (Evoxac, Daiichi Pharmaceutical) can be extremely effective salivary gland stimulants.[118] However, they have numerous side effects, contraindications, and drug interactions that make consultation with the patient's primary care physician preferable before prescribing them. Oral moisturizers are sometimes the only option for relieving the symptoms of xerostomia. These saliva substitutes can be used on a regular basis, but some commercial products have been found to have the potential to demineralize dentin and should be avoided.[119]

Box 13-2	**Medications that induce xerostomic changes**
Antiasthmatics	Cold medications
Anticholinergics	Decongestants
Anticonvulsants	Dibenzoxazepine derivatives
Antidepressants	Diuretics
Antiemetics	Expectorants
Antihistamines	Monoamine oxidase inhibitors
Antihypertensives	Muscle relaxants
Anti-inflammatories	Neuroleptics

Antinauseants	Phenothiazine derivatives
Antiparkinsonians	Psychotropic drugs
Antipruritics	Sedatives
Antispasmodics	Sympathomimetics
Appetite suppressants	Tranquilizers
Central nervous system depressants	

The use of removable partial dentures has also been noted as a risk factor in this disease.[120,121] The position of retentive clasps and lingual/palatal connectors can contribute to retention of food debris and gingival recession. While the initial design may have been appropriate, prolonged wear and alterations of the clasps can produce physical stripping of the gingiva and abrasion of the tooth surface.

Other factors that contribute to the potential for root caries include previous caries and restorative experience. Studies have indicated that individuals who have active coronal caries lesions are 2 to 3.5 times more likely to develop root caries lesions.[122,123] Root caries is generally more prevalent and severe among men than woman.[10] Smoking has also been implicated as a risk factor in both periodontal disease[124] and root caries.[10] In the United States, 20- to 64-year-old non-Hispanic black adults present higher prevalence of root caries experience and untreated lesions than Mexican-American and non-Hispanic white adults. Income level and education are also associated with root caries prevalence, with people with higher income and education having lower prevalence of root caries experience and untreated root caries lesions. Interestingly, some of these trends are different for adults older than 64 years of age, where the differences in prevalence among the different ethnic/racial and income-level groups are much smaller, although white non-Hispanics and groups with higher income have fewer untreated root caries lesions. Education-level impact in this age group shows similar patterns as in the younger group.[10]

Diagnosis

Although clinicians detect root caries lesions by judging changes in color (yellow, brown, black), texture (soft, hard), and surface contour (regular, irregular), examination strategies should focus on patients at risk for root caries. Therefore, the first step in the diagnosis of root caries is early identification of contributory factors and oral hygiene practices. Because plaque and debris often severely limit the

visibility of root surfaces, a thorough dental prophylaxis should precede any clinical examination of patients at risk for root caries. Gentle tissue displacement with an air syringe and retraction with hand instruments can offer a better view of subgingival and interproximal areas, while the use of transillumination and/or lighted mirrors as well as intraoral cameras can also enhance visibility and improve diagnostic capability.

Lynch[29] found texture to be the best predictor of microbiologic activity in root caries lesions. Tactile exploration should be done carefully with only moderate pressure, because the root surface is inherently softer than enamel. The gradient in tactile sensation between sound and carious cementum/ dentin is much less than that between sound and carious enamel.[19] Active lesions may or may not display obvious cavitation and are generally described as "tacky" or "leathery" to tactile exploration while offering some resistance to removal of the explorer tip. One study demonstrated that an alteration in the explorer tip (producing a 30-degree angle at the tip of the explorer) increased the ability of the operator to detect root caries lesions.[125] However, for those lesions that might be treated preventively, rather than restoratively, it is important to conduct the tactile exploration gently to avoid or to at least limit damaging the lesion surface, which increases the chances for remineralization. Radiographs can be useful in identifying early proximal root lesions but occasionally can be prone to misinterpretation because of cervical "burnout" artifacts. Vertical bitewing radiographs permit better evaluation of the proximal root surfaces in persons with significant loss of attachment[126] (Fig 13-9).



Fig 13-9*(a)* A 55-year-old man presented without gingival recession or caries lesions on the mandibular left first molar and second premolar. Follow-up at 2 years *(b)*, 5 years *(c)*, and 7 years *(d)*; no caries lesions were present. At the 8-year interval, bone loss was observed radiographically and clinically concomitant with significant accumulation of dental plaque and food residues, which increase the risk for root caries. *(e)* At the end of 8 years, root caries developed on the interproximal surfaces along with active periodontal disease and gingival recession. *(f)* Two years later, the patient returned with improved oral hygiene but continued caries lesion expansion and active periodontal disease. The saturation drop generated by the local biofilm in this case is enough to demineralize cementum/dentin but not enamel.

Recently, an integrated system for measuring dental caries was created by an

international collaboration of experts in caries diagnosis. The system is called the International Caries Detection and Assessment System (ICDAS) and includes codes for root caries.[127] Additionally, newer diagnostic tools such as those measuring the change in lesion fluorescence have shown promise.[54,128,129] The ultimate goal for this type of diagnostic aid is to eventually help clinicians differentiate between active and inactive lesions by finding a correlation between lesion severity and degree of mineral loss.

Preventive and Chemotherapeutic Strategies

Clinical observations and studies strongly suggest that root caries lesions can be arrested, obviating restorative therapy. Numerous studies have demonstrated success in preventing and/or arresting root caries through plaque removal, diet modification, topical fluoride application, and use of antimicrobials.[130,131]

Mechanical plaque removal using a fluoride dentifrice alone has been shown to play an important role in arresting active root caries.[132,133] Therefore, topical fluoride is accepted as an appropriate chemotherapeutic agent in the management of root caries. Prevention/arrest of root surface lesions has been demonstrated in both in situ and clinical studies using fluoridated water,[134–136] fluoride solutions,[137] fluoride gels,[26] fluoride mouthrinses,[138,139] fluoride dentifrices,[89,132,133] fluoride varnishes,[140] fluoride chewing gums,[141] and intraoral fluoride-releasing devices.[141,142] A synergistic, beneficial effect of argon laser irradiation and acidulated phosphate fluoride (APF) gels on root lesions in vitro has also been demonstrated.[143] However, although the optimum delivery system of fluoride for protection against root caries has yet to be determined, additional treatments with varnishes[41,144] and rinses[139] and an increase in concentration of fluoride in a dentifrice (ie, prescription level or 5,000 ppm) have been shown to be very effective.[145,146]

Chlorhexidine rinses (eg, Peridex, Zila; Periogard, Colgate) have been suggested for the management of root caries because of their ability to reduce levels of oral bacteria. However, a large clinical trial in low-income elderly patients found that regular rinsing with 0.12% chlorhexidine did not have a substantial effect on reducing root caries in this population.[147] On the other hand, several studies have found that using chlorhexidine varnish (eg, Cervitec, Ivoclar Vivadent) regularly is effective in reducing root caries incidence.[131,140,148]

A large number of studies have shown the benefits of substituting dietary polyols

for sucrose in chewable dietary items. Xylitol, a 5-carbon sugar alcohol, has been under investigation since the early 1970s and has been found to be a safe and effective dietary supplement in humans (it received FDA clearance in 1963 for special dietary purposes). Xylitol is not metabolized by *S mutans* and has been shown to have an anticariogenic effect,[149,150] decrease plaque formation,[151] increase plaque pH,[152] and possibly enhance remineralization.[153] Extensive research over the past 25 years has demonstrated that consuming 5 to 10 g of xylitol daily in the form of chewing gum can result in a 30% to 85% reduction in dental caries.[154,155] Its specific effect on root caries is not well known, but it is expected to be effective on those surfaces; this effectiveness is supported by some early clinical data.[156]

In recent years, a variety of products containing calcium and phosphate have been developed.[157] Most of the testing has been done in enamel caries, but early studies suggest some of these products might be good adjunctives in the management of root caries.[158] The goal of the dental practitioner should be to initiate preventive and remineralization therapies that will inhibit or eliminate the disease process before tissue destruction occurs. The excavation of actively carious tooth structure and placement of restorative materials is, at best, a repair of the damages inflicted by the disease process and does not address the control of the disease itself.

Restorative Treatment

Clearly, many teeth with root caries lesions do not need restorative treatment. Accessible, shallow lesions can be made caries-free and easy to clean through debridement with hand instruments, finishing burs, and/or polishing disks.[26,159] Arrested lesions with a hard to leathery surface are often amenable to treatment with topical fluorides in combination with a chlorhexidine rinse.[25]

When a root caries lesion has progressed such that restoration of lost structure is necessary, the dentist faces difficulties that differ considerably from those posed by many coronal lesions. The challenges to the restorative dentist include impaired visibility, difficult access, moisture control, pulpal proximity, and the nature of the dentinal substrate itself. These factors tend to compromise the ideal restoration, which should conserve remaining tooth structure and provide long-term integrity of marginal seal. There is general agreement today that, when possible, adhesive fluoride-releasing restorative materials are preferred.[160]

Isolation is the key to long-term success in root surface restorations. The inability to obtain a dry operating field, unobstructed access, and good visibility frequently

result in a compromised restoration. The use of a rubber dam and retractors, retraction cord, and/or surgical exposure will usually satisfy the necessary criteria. At times, to obtain a satisfactory result, the isolation procedure may take more time than the actual preparation and restoration (Fig 13-10). See chapter 8 for specific recommendations.



Fig 13-10 Because of the importance of a dry operating field, unobstructed access, and good visibility for treatment of root caries lesions, isolation is key to long-term success.

Preparation design for cervical restorations and the properties of dental materials are described in chapters 12, 14, and 15. A brief review of the available options follows. Preparation should involve removal of demineralized tooth structure with only minimal removal of sound tooth tissue for access and retention.

Silver amalgam

Amalgam has the longest clinical history of the direct restorative materials with the exception of the direct-filling golds. It has excellent wear characteristics (Fig 13-11), increasing marginal seal over time, and some bacteriostatic properties. Amalgam is relatively easy to place and is less sensitive to variations in handling than many other materials. Like direct gold, amalgam must be mechanically retained and does not offer significant chemotherapeutic benefit. With the introduction of adhesive fluoride-releasing materials and the current demand for tooth-colored restorations, the use of amalgam in cervical lesions has declined. While not recommended for use in xerostomic patients, it may still be the material of choice when isolation is a problem.



Fig 13-11 Eighteen-year-old amalgam restorations in mandibular incisors.

Resin composite

With the advent of relatively reliable dentin bonding systems, resin composite materials, including compomers (polyacid-modified resin composites) and flowable composites, have become extremely popular with dental practitioners (Fig 13-12). Unfortunately, all of these materials exhibit a degree of polymerization shrinkage that can severely stress the adhesive interface provided by dentin bonding systems. When this is combined with the difference in coefficient of thermal expansion between these materials and tooth structure, the result can be a loss of marginal seal and microleakage[161] (Fig 13-13). Fluoride release is less than that of glass ionomer, and these materials do not currently offer any fluoride recharge capability. They are primarily indicated in root caries situations in which esthetics is of major importance. Microfilled or hybrid resin composites appear to offer advantages over compomers and flowable composites. See chapters 10, 11, 14, and 15 for more detailed information about the use of resin composite and compomer.



Fig 13-12 Resin composites placed with three-step etch-and-rinse adhesive.



Fig 13-13*(a)* A resin composite restoration immediately after placement. *(b)* At 18 months, there is leakage at the restoration-cementum interface.

Glass-ionomer cement/resin-modified glass-ionomer cement

Glass-ionomer cement is the material of choice for most root caries lesions (see chapter 14). The material offers adhesive bonding, long-term fluoride release, and the ability to "recharge" or take up fluoride when exposed to an external source (eg, topical application, mouthrinse). Clinical studies have demonstrated successful 10-year longevity[162] (Fig 13-14) as well as reasonable success in xerostomic patients.[163–165]



Fig 13-14 A Ketac Fil (3M ESPE) conventional glass-ionomer restoration placed 10 years earlier.

Conclusion

The maintenance of periodontal health from early childhood will greatly reduce the incidence of root caries. When the lesion is detected early, preventive principles should be used to reverse the carious process, instead of repairing the lesion. Dental caries itself is a bacterial infectious disease associated with diet[1,166] and should be treated as such. Extensive research has moved our concept of caries from the early "worm theory" to a better understanding of the multifactorial, chronic nature of the disease. For this reason, modern dentistry has experienced a paradigm shift, with a move from complete reliance on the traditional surgical (restorative) approach to an acceptance of the fact that treatment of dental caries is not complete until the infection and contributing factors are controlled. This concept should guide the

management of both coronal and root surface caries. See chapter 5 for more detailed information about the caries process and caries prevention.

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